Prompt: Consider the following numbered solutions to this problem:

"At $t=0$, a resistor (resistance R=10 Ohms) is connected across a fully charged capacitor (voltage V=5V, capacitance C=100mF). At what time t_2 does half the initial current flow through the resistor?"

Grade them from 0 (worst) to 4 (best) in terms of each correctness of argumentation, symbolic correctness, numerical correctness, and straightforwardness. Provide these grades in a comma-separated table with an additional column that gives a one-sentence feedback. Also, provide a comma-separated matrix of similarity between the solutions based on approaches, with the solution number as header row.

======== Solution 1:
We have a resistor (R = 10 Ohms) connected across a fully charged capacitor (V = 5V, C = 100mF).
...
Therefore, the time t_2 when half the initial current flows through the resistor is approximately $0.693,\text{s}$.
...
======== Solution 22:
The time constant τ of an RC circuit is given by $\tau = RC$. In this case, $\tau = RC = (10\text{ Ohms})(100\text{ mF}) = 1\text{ s}$.
...

FIG. 8. Example of a grading prompt.

Prompt: The following are feedbacks on solutions to problem 1 on a test, prepared by different graders. Summarize these statements into one feedback reflecting the majority of the graders.
* Solution 1 is correct and straightforward.
* Correct and detailed solution.
* The solution is correct and straightforward, using the correct formula for the current in an RC circuit and solving for the required time.
* Correct and straightforward approach with correct calculations and explanation.
* A complete and accurate solution using the correct approach.
...

FIG. 9. Example of a prompt to summarize the grading feedback.

IV. RESULTS

A. Grading Results

Tables I, II and III show the rubric-grading results by GPT-4, based on an average of 15 rounds of grading within different batches of five problems each.

Striking are the scores that have more than one point of standard deviations to their averages, which are:

- The correctness of the argument in Solutions 10, 15, 21, 23, 24, and 25: All of these solutions in the end arrive at the wrong numerical answer. Solutions 10 and 15 also needlessly calculate a numerical value for the initial current, while Solutions 21, 23, and 25 take a slight detour via the capacitative current $I(t) = C \cdot dV/dt$. Solution 24 starts with the expression for a charging instead of a discharging capacitor, which would still have arrived at the same answer had it not been for numerical errors.

- The correctness of the symbolic calculations in Solutions 15 and 23: Solution 15 is plug-and-chug, i.e., it plugs numbers into the first available formula and then transfers the numerical result to the next formula. Thus, the score for symbolic correctness of Solution 15 should have been consistently low for all grading rounds. Solution 23 makes an error in the very first line of the symbolic calculations by not eliminating the capacitance C.

- The correctness of the numerical results, in particular for Solutions 2, 3, 4, 6, 7, etc. — 18 out of the 25 problems altogether: 8 of these solutions actually have the correct numerical answer, 4 are wrong by one or more orders of magnitude, and 6 of the problems have a completely different numerical answers due to various reasons. The large list may be the result of GPT 4 still falling short on numerical calculations.

- The straightforwardness of Solutions 15, 21, 23, and 25: All of these solutions already appeared in other lists of ambiguous scores.

Overall, there does not appear to be a discernible pattern as to why these solutions are leading to widely spread scores. The one-sentence feedback given by GPT-4 also does not provide helpful hints for determining which features of these solutions the algorithm may have latched on to.

The summaries of the one-sentence feedback message sound very plausible, but many of them include incorrect or at least misleading statements. Examples of incorrect statements include:

- "The majority opinion indicates that there is an error in the calculation of the time constant, leading to an incorrect numerical result for t_2" for Solution 2, when in fact the numerical result is correct.

- "The majority of graders agree that Solution 24 uses the correct approach, equations, and symbolic representation," when in fact the equation for a charging capacitor is used.

Examples of misleading statements are:

- "One grader notes that the final answer should be in milliseconds, not seconds" for Solution 1; while it is true that microseconds would have been more elegant, this is not required.

- "Solution 17 demonstrates the correct approach," when it fact it includes unnecessary calculations.

TABLE I. AI-grading of Solutions 1–10.

Sol.	Arg.	Symb.	Num.	Str.forw.	Feedback
1	4.0 ± 0.0	4.0 ± 0.0	3.9 ± 0.3	4.0 ± 0.0	The majority of graders agree that Solution 1 is correct, straightforward, and well-structured. The solution uses the correct formula for the current in an RC circuit, derives the expression for the time t_2, and solves for the required time when half the initial current flows through the resistor. One grader notes that the final answer should be in milliseconds, not seconds. Overall, the solution is clear, concise, and logically explained.
2	3.6 ± 0.9	3.7 ± 0.6	2.8 ± 1.6	3.8 ± 0.4	The majority of graders agree that the solution to problem 2 has a correct and clear approach, using the appropriate RC circuit equations and argumentation. However, there seems to be some disagreement on the numerical calculation for t_2. The majority opinion indicates that there is an error in the calculation of the time constant, leading to an incorrect numerical result for t_2. Overall, the solution is well-organized and logically explained, but the numerical value should be revised.
3	3.7 ± 0.5	3.7 ± 0.5	2.2 ± 1.6	3.7 ± 0.5	The majority of graders agree that Solution 3 has a correct and straightforward approach, with correct argumentation and symbolic representation. However, there is a common issue with the numerical calculation for t_2, with some graders indicating the result is off by a factor of 1000 or in the wrong unit (milliseconds instead of seconds). Despite these errors, the solution is generally well-explained and follows a logical approach.
4	3.7 ± 0.5	3.8 ± 0.4	3.3 ± 1.2	3.8 ± 0.4	Solution 4 demonstrates a correct and straightforward approach to solving problem 4, using appropriate RC circuit equations and accurately deriving the time t_2. However, the majority of the graders point out a minor numerical error in the final answer, possibly due to rounding or unit conversion. Despite this, the solution is clear, well-argued, and symbolically correct.
5	3.9 ± 0.4	3.9 ± 0.4	3.9 ± 0.4	3.9 ± 0.4	Solution 5 is generally correct and straightforward, successfully applying the exponential decay equation for the RC circuit to derive the result for t_2. The majority of the graders appreciate the clear argumentation, symbolic correctness, and numerical accuracy. However, there is a minor numerical error due to rounding, and some graders suggest that the clarity and straightforwardness of the argument could be improved. Overall, the solution demonstrates a good understanding of Ohm's law, the time constant, and the current equation for an RC circuit.
6	3.8 ± 0.4	3.8 ± 0.4	3.3 ± 1.1	3.8 ± 0.4	The majority of the graders agree that the solution to problem 6 is correct, well-argued, and straightforward in its approach, with accurate symbolic representation and explanation. However, some graders noted an incorrect numerical value in the final result or a mistake in calculating t_2.
7	3.1 ± 0.9	3.6 ± 0.6	3.1 ± 1.1	3.1 ± 0.9	The majority of the graders found Solution 7 to be correct with accurate argumentation, symbolic representation, and numerical values. However, some graders mentioned issues such as incorrect current direction, voltage calculation, and final result, as well as an incorrect manipulation of the equation. Despite these discrepancies, most graders agreed that the solution was straightforward and used the correct approach.
8	3.8 ± 0.4	3.8 ± 0.4	3.4 ± 1.1	3.8 ± 0.4	The majority of graders agree that Solution 8 has correct argumentation, symbolic representation, and a straightforward explanation. The solution correctly derives the formula for the time at which half the initial current flows and uses the appropriate approach. However, there is disagreement on the numerical value, with some graders noting that it is incorrect or off by a factor due to not considering units. Overall, the solution is mostly correct, but the final numerical answer may need to be revised.
9	0.2 ± 0.4	1.0 ± 0.9	0.0 ± 0.0	0.8 ± 0.7	Solution 9 demonstrates an incorrect understanding of the initial current, mistakenly assuming it to be zero. This leads to incorrect calculations, argumentation, and the false conclusion that half the initial current never flows through the resistor. The overall approach and reasoning are flawed, resulting in an incorrect solution.
10	2.8 ± 1.1	3.1 ± 0.8	1.3 ± 1.3	2.9 ± 0.9	The majority of the graders agree that the solution to problem 10 has correct argumentation and symbolic representation, with a clear and straightforward approach. However, there are errors in the numerical calculations, including the time constant and natural logarithm term, leading to an incorrect final result. Some graders also mentioned unnecessary complexity and incorrect derivations.

TABLE II. AI-grading of Solutions 11–20.

Sol.	Arg.	Symb.	Num.	Str.forw.	Feedback
11	3.5 ± 0.8	3.5 ± 0.7	2.8 ± 1.4	3.5 ± 0.8	The majority of the graders agree that Solution 11 has a correct approach by using the exponential decay equation for the RC circuit and deriving the equation for i(t). However, there seems to be a consensus that there are errors in the calculations, leading to an incorrect numerical result for t_2. The argumentation, symbolic correctness, and straightforwardness could be improved. Despite these errors, some graders still consider the solution to be well-structured and clear.
12	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	Solution 12 is a correct, well-structured, and straightforward approach to the problem. The majority of graders praised the clear argumentation, appropriate use of symbols and equations, and accurate numerical results. The solution effectively uses the RC circuit formula and provides a concise explanation.
13	3.9 ± 0.3	3.9 ± 0.3	3.5 ± 1.0	3.9 ± 0.3	The majority of the graders agree that Solution 13 is correct, clear, and straightforward. The solution uses the appropriate RC circuit equations, symbols, and approach to find the time when half the initial current flows through the resistor. While there is mention of a sign error and incorrect final result by a couple of graders, the overall consensus supports the solution's correctness and organization.
14	3.2 ± 0.9	3.4 ± 0.8	2.3 ± 1.4	3.2 ± 0.9	The majority of graders agree that Solution 14 has a correct approach and argumentation, with symbolic correctness and straightforwardness. However, there are inconsistencies in the numerical results, likely due to errors in calculations or sign errors.
15	2.4 ± 1.4	2.4 ± 1.4	1.9 ± 1.6	2.3 ± 1.5	The majority of graders agree that the solution to problem 15 is mostly accurate in terms of approach and equations used, but there are some minor mistakes in argumentation, symbolic correctness, and numerical calculations. The solution is considered to be somewhat straightforward and well-explained, but with a few errors in derivation and numerical results.
16	3.3 ± 0.8	3.6 ± 0.5	1.9 ± 1.5	3.6 ± 0.5	The majority of the graders agree that Solution 16 has a correct approach and demonstrates accurate symbolic representation, argumentation, and use of formulas. However, a calculation mistake in determining the initial current and subsequent errors in numerical calculations lead to an incorrect numerical answer for t_2. Despite these numerical issues, the solution is well-organized and clear.
17	3.3 ± 1.0	3.8 ± 0.6	2.2 ± 1.4	3.6 ± 0.6	Solution 17 demonstrates the correct approach, argumentation, and symbolic representation, but makes an error in the numerical calculation, specifically a sign error in the time constant, which leads to an incorrect final result. Despite this, the solution is considered straightforward and directly addresses the problem.
18	3.1 ± 0.8	3.4 ± 0.6	2.6 ± 1.2	3.2 ± 0.7	Solution 18 has the correct approach and argumentation, but the majority of graders point out an incorrect time constant calculation, which leads to an incorrect final numerical result. Despite this, the solution is well-structured and uses proper symbols and numerical values.
19	0.3 ± 0.6	0.9 ± 0.9	0.2 ± 0.4	0.5 ± 0.7	The majority of the graders agree that Solution 19 contains multiple errors, including incorrect initial current and voltage calculations, as well as an incorrect approach for finding the time t_2 when half of the initial current flows through the resistor. These errors lead to an incorrect final result. The solution also has issues with argumentation, symbolic representation, and straightforwardness.
20	3.1 ± 0.7	3.5 ± 0.5	2.0 ± 1.3	3.3 ± 0.6	Solution 20 generally employs the correct approach and uses the exponential decay equation for solving the problem, but there are errors in the numerical calculations, specifically in the time constant and initial current. These inaccuracies lead to an incorrect final result. While some graders appreciate the clear and straightforward steps, the majority point out the numerical errors affecting the outcome.

TABLE III. AI-grading of Solutions 20–25.

Sol.	Arg.	Symb.	Num.	Str.forw.	Feedback
21	1.6 ± 1.1	1.9 ± 1.0	0.7 ± 0.8	1.8 ± 1.1	The majority of the graders agree that Solution 21 has an incorrect approach, incorrect or incomplete argumentation, and incorrect numerical calculations. The relationship between i(t) and dv/dt is also incorrect, leading to a wrong calculation for t_2. Some graders acknowledge clear explanation, symbolism, or straightforwardness in parts of the solution, but overall, the solution is deemed incorrect and weak.
22	3.7 ± 0.6	3.9 ± 0.4	3.4 ± 1.2	3.7 ± 0.6	The majority of graders agree that Solution 22 is correct and straightforward in its approach to finding the time when half the initial current flows through the resistor. The solution uses the time constant and the voltage across the capacitor as a function of time. However, a few graders noted some minor issues with the numerical calculation and representation of the time constant. Overall, the solution is well-explained and clear, with accurate derivation and symbolic correctness.
23	2.1 ± 1.5	2.6 ± 1.1	1.0 ± 1.6	2.1 ± 1.4	The majority of the graders find Solution 23 to have a correct approach and clear step-by-step argumentation. However, there are concerns about the incorrect final numerical result and potential errors in the calculations involving the exponential term and natural logarithm. Some graders also mention an incorrect use of the time constant, incorrect equation, and wrong numerical value.
24	2.9 ± 1.2	3.5 ± 0.7	1.7 ± 1.4	3.0 ± 0.9	The majority of graders agree that Solution 24 uses the correct approach, equations, and symbolic representation to find the time at which half the initial current flows through the resistor. However, there are numerical errors in the calculation, particularly regarding the time constant and the voltage across the capacitor. Additionally, some graders noted unnecessary complexity in the solution. Overall, the solution is partially correct but suffers from incorrect numerical values and minor inconsistencies.
25	1.7 ± 1.3	2.7 ± 0.8	0.7 ± 1.1	2.1 ± 1.1	The majority of the graders agree that Solution 25 has an incorrect numerical result for t_2 due to inconsistencies, errors, and incorrect values in the derivation and equation. The approach is mostly correct and some steps are symbolically accurate, but there are sign errors and the method is not straightforward.

Often this includes remarks about numerical inaccuracies, even if the result is correct. However, particularly for the solution attempts that were completely incorrect, the feedback can be useful, for example, "Solution 9 demonstrates an incorrect understanding of the initial current, mistakenly assuming it to be zero."

B. Agreement with Manual Grading

Fig. 10 shows the correlation between the rubric scores resulting from manual and AI-grading. The scores are clearly positively correlated, with a particularly high R^2 for the correctness of the argument and the numerical answer. This finding is surprising, since the correctness of the argument is more subjective than for example the correctness of the symbolic operations, and the scoring of the correctness of the numerical answers is the one with the highest standard deviation between grading cycles. Symbolic correctness has the lowest R^2 and is generally rated a lot higher by the AI than by the author.

Fig. 11 shows the total scores on the problem, where the rubric items approach, symbolic correctness, and straightforwardness are weighted 20%, and the correct final numerical answer 40%, respectively. In addition, the correspondingly combined standard deviations for the AI-gradings are given by error bars, and the data

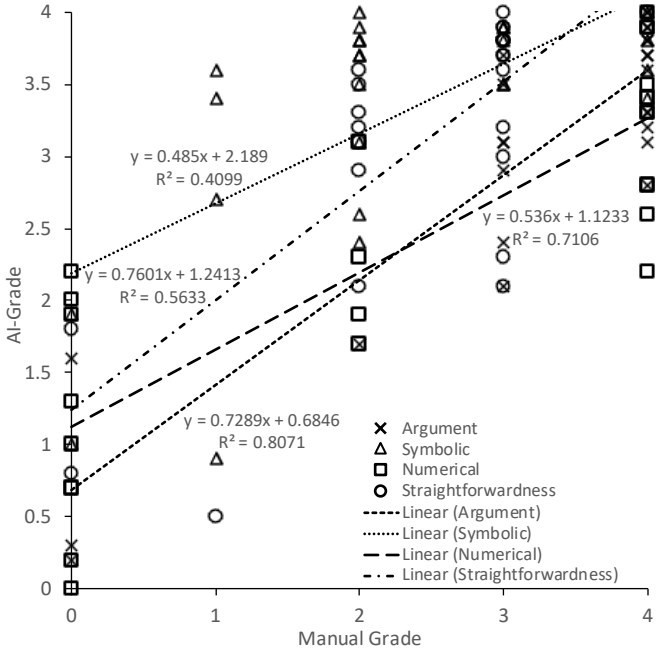

FIG. 10. Correlation between manual and AI-grading rubric scores.

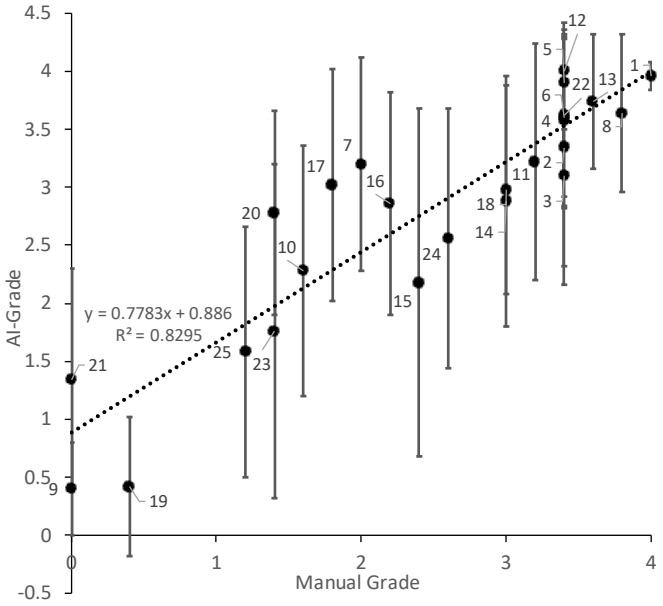

FIG. 11. Correlation between manual and AI-grading total scores. For each data point, the solution number and combined AI-grading standard deviations are indicated.

points are labeled by solution number.

The scores for the best solutions agree fairly well with the linear interpolation ($R^2 \approx 0.83$), but fluctuations are higher in partial credit situations. On the average, GPT assigns total scores that are almost 0.9 points higher than those by the author

C. Clustering

Figure 12 shows dendrograms and heat maps of the similarities between the solutions based on manual grading, AI-grading, and AI-approach similarity, respectively. Based on these, three clusters emerge for the manual and the AI-grading. For the AI-similarity-of-approach measure (determined by GPT in response to the prompt in Fig. 8), either two clusters could be identified, or the dendrogram could be cut at a deeper level (indicated by dashed lines in Fig. 12), resulting in four clusters. However, the fourth cluster only contains Solution 23, so a decision to not treat it separately could be justified.

These clusters are shown in Table. IV; it turns out that they roughly align with the total points in Fig. 11, with the low, medium, and highly scored problems forming the clusters.

All similarity measures identify the almost entirely wrong solutions 9, 19, and 21 as members of the "low" cluster. The manual grading has Solution 23 in the middle cluster, as the only mistake is $\frac{CV_0}{-RC} = \frac{V_0}{RC}$, that is, an error in the calculation rather than a fundamental error; this solution was singled out in the clustering according to similarity-of-approach. Solution 25 not once but twice

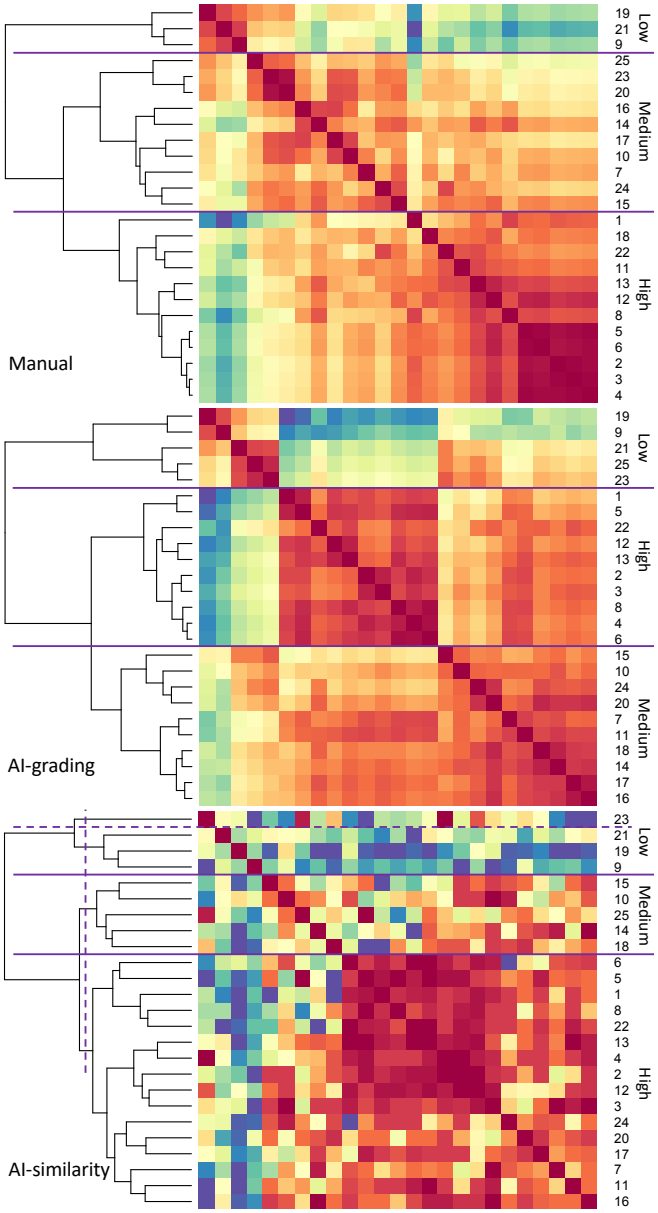

FIG. 12. Dendrograms and heat maps of similarities between solutions based on manual grading (top panel), AI-grading (middle panel), and AI-approach similarity (bottom panel). The cluster cuts are indicated by purples lines; the dashed line indicates possible cuts.

includes the same kind of error in calculating fractions.

Solutions 11 and 18 made it into the highest cluster for manual grading, but are not found in the corresponding cluster for AI-grading. Both solutions arrive at the correct result (even though Solution 18 is somewhat nonchalant with the units in the second-to-last expression).

Overall, clustering the solutions based on the similarity of the scores within the grading rubric provided similar information for manual and AI-grading, while clustering based on the similarity-of-approach is less informative.

TABLE IV. Clustering based on the dendrograms and heat maps in Fig. 12.

	Low	Medium	High
Manual	9 19 21	7 10 14 15 16 17 20 23 24 25	1 2 3 4 5 6 8 11 12 13 18 22
AI-grading	9 19 21 23 25	7 10 11 14 15 16 17 18 20 24	1 2 3 4 5 6 8 12 13 22
AI-similarity	9 19 21 (23)	10 14 15 18 25	1 2 3 4 5 6 7 8 11 12 13 16 17 20 22 24

The automatically formed clusters based on AI could be used for sorting solutions in preparation for human grading.

V. DISCUSSION

A. Potential for Assisting in Grading

GPT shows great promise as a tool for grading written solutions to physics problems, as evidenced by the results in Fig. 11. The system performs best at the extreme ends of the grading spectrum: clearly correct and clearly incorrect solutions are generally reliably recognized, but overall, the system currently falls short of the stringency required for standalone, unsupervised exam grading. Also, the system generally assigns more points than the author would.

While there are proposals to mitigate these shortcomings by providing "high transparency" assessments with multiple attempts and regrade requests [30], students will likely only invoke these for false negatives, i.e., scores that are lower than what they expected. This could lead to grade inflation. Instead of waiting for student complaints, instructional staff will need to verify the scores before the exam is returned.

As typical for the tool, feedback sounds plausible, but might be misleading. In grading problems, GPT-4 still exhibits some of the same weaknesses with symbolic and numerical calculations that its predecessor GPT-3 did in solving problems [6]. OpenAI promises the optional incorporation of Wolfram Alpha [31], which may solve this issue, but just like for the API, there is also a waiting list for the required plugin.

B. Potential for Assisting in Learning

If not used for grading exams, especially in connection with smartphone-based scanning, GPT or similar tools could be used for assisting students in learning. This is a far more classic domain for AI in physics education [32, 33]

In that context it is important if the system could pinpoint particular errors. At the moment, though, this is still somewhat doubtful. For example, when prompted to find the mistake in Solution 14, the system responds:

> The mistake in the solution is in the definition of the time constant. The correct definition is $RC = R \cdot C$. The rest of the solution is correct. We need to correct the time constant and then find the correct value of t_2.
>
> The time constant is defined as $RC = R \cdot C$. In this case, $R = 10, \text{Ohms}$ and $C = 100, \text{mF} = 100 \cdot 10^{-3}, \text{F}$, so the time constant is:
>
> $RC = R \cdot C = 10, \text{Ohms} \cdot 100 \cdot 10^{-3}, \text{F} = 1, \text{s}$
>
> Now we can substitute the corrected time constant into the equation for t_2:
>
> $t_2 = -RC \ln \frac{1}{2} = -1, \text{s} \ln \frac{1}{2} \approx 0.693, \text{s}$
>
> So the correct time t_2 at which half the initial current flows through the resistor is approximately 0.693 seconds.

The error is not really in the "definition" of the time constant, but simply in the calculation of the powers of ten.

When prompted to pinpoint the error in Solution 23, the response is:

> The mistake in the solution is in the derivation of the equation $i(t_2) = \frac{1}{2} i(0)$. The correct equation should be:
>
> $i(t_2) = \frac{1}{2} i(0) = \frac{1}{2} \frac{V_0}{R}$
>
> However, the solution incorrectly equates this expression to the equation for $i(t)$:
>
> $-\frac{V_0}{2RC} e^{-\frac{t_2}{RC}}$
>
> The correct equation for $i(t)$ should be:
>
> $i(t) = \frac{V_0}{R} e^{-\frac{t}{RC}}$
>
> Now, we can set up the correct equation for $i(t_2)$:
>
> $\frac{1}{2} i(0) = \frac{1}{2} \frac{V_0}{R} = \frac{V_0}{R} e^{-\frac{t_2}{RC}}$
>
> Now, we solve for t_2:
>
> $e^{-\frac{t_2}{RC}} = \frac{1}{2}$
>
> $-\frac{t_2}{RC} = \ln\left(\frac{1}{2}\right)$
>
> $t_2 = RC \ln 2 = 10\text{s} \ln 2 \approx 6.93\text{s}$
>
> So the time t_2 at which half the initial current flows through the resistor is approximately 6.93 seconds.

The system correctly found that the C was not cancelled, but then actually proceeded to wrongly calculate the answer. The feedback given by the system can be helpful,

but needs to be evaluated critically by the learner — which in itself is an increasingly valuable competency, as AI will almost invariably become ubiquitous in everyday life.

In any case, the system should not attempt to give away the solution; unfortunately, using the prompt "Act as a socratic teacher and assist a learner with this solution attempt" made the reply more personable and supportive ("You're on the right track, but there is a small mistake ..."), but did not stop the system from providing a solution, either.

VI. LIMITATIONS

This study is limited to one particular tool, GPT-4, at one particular point in time, mid-April 2023. Due to the rapid development of AI-tools, it can thus provide nothing more than a snapshot of what is minimally possible. The initial step, conversion of handwritten solutions involving mathematical expressions to a machine-readable format could not be investigated here, due to lack of access to the required Application Programming Interface.

The author had to make choices when providing prompts to GPT. While some experimentation went into formulating the prompts, due to the probabilistic nature of the responses, some decisions were made based on anecdotal evidence, and better prompts framing the solutions could well have resulted in more reliable results.

GPT is only one of the rapidly evolving tools becoming broadly available. There are competing solutions by Google [34], NVIDIA [35], and Microsoft [36], which may perform better or worse, but could not be evaluated here.

VII. OUTLOOK

The next step to this exploratory study would be enabled by access to the API (either of GPT or of other tools on the market) and consist of fully implementing the workflow in Fig. 1; performance of the optical character recognition is also crucial to the reliability of future systems.

A future study should involve authentic student work instead of GPT grading itself, for example from exams in large-enrollment physics courses, and compare the results from AI-grading to those from a traditional grader pool. Particularly in high-stake exams, grading usually involves more than one round, so inter-rater reliability could also be established for the human graders. As a welcome side-effect, earlier steps in Fig. 1 could provide better workflow management even for human graders, as less paper would be shuffled around and grades could more easily be adjusted.

VIII. CONCLUSION

In this exploratory study, GPT has shown considerable potential for grading freeform student work in physics. While AI-assigned grades have a strong correlation to manually assigned grades, they are currently not reliable enough for summative assessments, such as high-stake exams. The system, however, is reliable enough to assist human graders by pre-sorting or clustering solutions and by providing preliminary scores. GPT still remains hampered by its limited capabilities and inconsistencies carrying out symbolic and numerical calculations, so currently several independent grading rounds are needed. The narrative feedback provided by the system seems plausible, but currently still frequently falls short of being reliable. The system can be helpful in formative assessment, but also in that scenario, learners need to critically evaluate its responses.

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